STRUCTURAL CALCULATION NOTE

East Steel Frame ST-E2 with East Arch Hangers - stand-alone shopfit steel

Analysis and results exported from Autodesk Robot Structural Analysis Professional 2026

مشروع تصميم وتنفيذ أعمال المعرض المتحفي بمقر مؤسسة الملك عبدالله

Madina 2 Store (Madinah style), Knowledge Economic City

Project No. SIC-043-GEN-GSH2

CLIENT

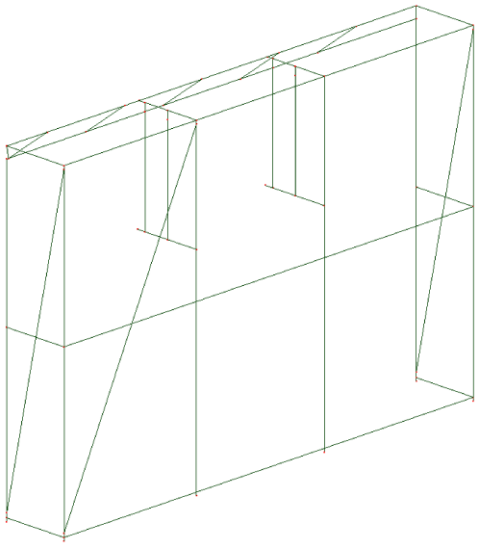
مؤسسة الملك عبدالله الإنسانية
King Abdullah Humanitarian Foundation
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Hadaya Taiba Co.

Document No.	MD2-SF-ZZ-GF-CA-S-0001
Revision	00 - for review
Date	12.11.2026
Design codes	SBC 301-2018 gravity and notional loads (indoor, no wind); SBC 306 / AISC 360-16 LRFD; SHS 40x40x2, L 40x40x4, M10 rods, flat bars 40x6 and 116x10
Robot model	Madina2_ST-E2_asbuilt.rtd
Prepared by	Samaya Factory - Technical Office
Checked by	Mohamed Sultan
Approved by	Client engineer (name to be given)

This document is the property of Samaya Factory. Preliminary issue for review; not for construction until approved.

Structure View



Data - Members

Member	Node 1	Node 2	Section	Material	Length (m)	Gamma (Deg)	Type
1	1	2	SHS 40x40x2	STEEL	0.03	0.0	Simple member
2	2	3	SHS 40x40x2	STEEL	0.04	0.0	Simple member
3	3	4	SHS 40x40x2	STEEL	1.43	0.0	Simple member
4	4	5	SHS 40x40x2	STEEL	1.30	0.0	Simple member
5	5	6	SHS 40x40x2	STEEL	0.02	0.0	Simple member
6	6	7	SHS 40x40x2	STEEL	0.10	0.0	Simple member
7	8	9	SHS 40x40x2	STEEL	0.03	0.0	Simple member

8	9	10	SHS 40x40x2	STEEL	0.04	0.0	Simple member
9	10	11	SHS 40x40x2	STEEL	1.43	0.0	Simple member
10	11	12	SHS 40x40x2	STEEL	1.30	0.0	Simple member
11	12	13	SHS 40x40x2	STEEL	0.10	0.0	Simple member
12	14	15	SHS 40x40x2	STEEL	0.03	0.0	Simple member
13	15	16	SHS 40x40x2	STEEL	0.03	0.0	Simple member
14	16	17	SHS 40x40x2	STEEL	1.44	0.0	Simple member
15	17	18	SHS 40x40x2	STEEL	1.37	0.0	Simple member
16	18	19	SHS 40x40x2	STEEL	0.03	0.0	Simple member
17	20	21	SHS 40x40x2	STEEL	1.90	0.0	Simple member
18	21	22	SHS 40x40x2	STEEL	0.97	0.0	Simple member
19	22	23	SHS 40x40x2	STEEL	0.03	0.0	Simple member
20	24	25	SHS 40x40x2	STEEL	1.90	0.0	Simple member
21	25	26	SHS 40x40x2	STEEL	1.00	0.0	Simple member
22	27	28	SHS 40x40x2	STEEL	0.03	0.0	Simple member
23	28	29	SHS 40x40x2	STEEL	1.47	0.0	Simple member
24	29	30	SHS 40x40x2	STEEL	1.37	0.0	Simple member
25	30	31	SHS 40x40x2	STEEL	0.03	0.0	Simple member
26	7	32	SHS 40x40x2	STEEL	0.42	0.0	Simple member
27	32	33	SHS 40x40x2	STEEL	0.79	0.0	Simple member
28	33	34	SHS 40x40x2	STEEL	0.15	0.0	Simple member
29	34	35	SHS	STEEL	0.64	0.0	Simple

			40x40x2				member
30	35	36	SHS 40x40x2	STEEL	0.67	0.0	Simple member
31	36	37	SHS 40x40x2	STEEL	0.12	0.0	Simple member
32	37	38	SHS 40x40x2	STEEL	0.79	0.0	Simple member
33	38	13	SHS 40x40x2	STEEL	0.62	0.0	Simple member
35	6	39	SHS 40x40x2	STEEL	0.80	0.0	Simple member
36	39	40	SHS 40x40x2	STEEL	0.79	0.0	Simple member
37	40	41	SHS 40x40x2	STEEL	0.79	0.0	Simple member
38	41	42	SHS 40x40x2	STEEL	0.79	0.0	Simple member
39	42	12	SHS 40x40x2	STEEL	1.01	0.0	Simple member
40	6	32	SHS 40x40x2	STEEL	0.41	0.0	Simple member
41	39	33	SHS 40x40x2	STEEL	0.41	0.0	Simple member
42	40	35	SHS 40x40x2	STEEL	0.41	0.0	Simple member
43	41	37	SHS 40x40x2	STEEL	0.41	0.0	Simple member
44	42	38	SHS 40x40x2	STEEL	0.41	0.0	Simple member
45	7	19	SHS 40x40x2	STEEL	0.59	0.0	Simple member
46	13	31	SHS 40x40x2	STEEL	0.59	0.0	Simple member
47	3	18	SHS 40x40x2	STEEL	2.86	0.0	Simple member
48	10	30	SHS 40x40x2	STEEL	2.86	0.0	Simple member
49	4	17	SHS 40x40x2	STEEL	0.59	0.0	Simple member
50	11	29	SHS 40x40x2	STEEL	0.59	0.0	Simple member
51	2	15	SHS 40x40x	STEEL	0.59	0.0	Simple member

			2				
52	9	28	SHS 40x40x 2	STEEL	0.59	0.0	Simple member
53	15	28	SHS 40x40x 2	STEEL	4.21	0.0	Simple member
54	17	29	SHS 40x40x 2	STEEL	4.21	0.0	Simple member
55	19	23	SHS 40x40x 2	STEEL	1.36	0.0	Simple member
56	23	26	SHS 40x40x 2	STEEL	1.32	0.0	Simple member
57	26	31	SHS 40x40x 2	STEEL	1.53	0.0	Simple member
58	16	22	SHS 40x40x 2	STEEL	3.12	0.0	Simple member
59	34	43	L 40x40x 4	STEEL	0.06	0.0	Simple member
60	43	44	L 40x40x 4	STEEL	0.23	0.0	Simple member
61	44	23	L 40x40x 4	STEEL	0.30	0.0	Simple member
62	43	45	M10 rod	STEEL	0.07	0.0	Simple member
63	45	46	FB 40x6	STEEL	0.93	0.0	Simple member
64	44	47	M10 rod	STEEL	0.07	0.0	Simple member
65	47	48	FB 40x6	STEEL	0.93	0.0	Simple member
66	21	48	PL 116x10	STEEL	0.30	0.0	Simple member
67	48	46	PL 116x10	STEEL	0.23	0.0	Simple member
68	46	49	PL 116x10	STEEL	0.08	0.0	Simple member
69	36	50	L 40x40x 4	STEEL	0.06	0.0	Simple member
70	50	51	L 40x40x 4	STEEL	0.23	0.0	Simple member
71	51	26	L 40x40x 4	STEEL	0.30	0.0	Simple member
72	50	52	M10 rod	STEEL	0.07	0.0	Simple member
73	52	53	FB 40x6	STEEL	0.93	0.0	Simple member
74	51	54	M10 rod	STEEL	0.07	0.0	Simple member
75	54	55	FB 40x6	STEEL	0.93	0.0	Simple member
76	25	55	PL 116x10	STEEL	0.30	0.0	Simple member

77	55	53	PL 116x10	STEEL	0.23	0.0	Simple member
78	53	56	PL 116x10	STEEL	0.08	0.0	Simple member

Data - Panels

Data - Sections

	Section name	Member list	AX (cm2)	AY (cm2)	AZ (cm2)	IX (cm4)	IY (cm4)	IZ (cm4)
	SHS 40x40x2	1to33 35to58	3.200	1.600	1.760	12.736	9.195	7.915
	PL 116x10	66to68 76to78	11.600	9.667	9.667	3.657	0.967	130.075
	L 40x40x4	59to61 69to71	0.0	0.0	0.0	0.0	0.0	0.0
	M10 rod	62 64 72 74	0.581	0.490	0.490	0.054	0.027	0.027
	FB 40x6	63 65 73 75	2.400	2.000	2.000	0.261	0.072	3.200

Data - Supports

	Support name	List of nodes	List of edges	List of objects	Support conditions
	Pinned	1 8 14 20 24 27			UX UY UZ

Loads - Cases

Case	Label	Case name	Nature	Analysis type

1	DL	DL	dead	Static - Linear
2	LL	LL	live	Static - Linear
3	H	H	live	Static - Linear
10		SBC 1.2D+1.6L	dead	Linear Combination
11		SBC SLS D+L	dead	Linear Combination
12		SBC 1.4D	dead	Linear Combination
13		SBC 0.9D+1.6L (uplift)	dead	Linear Combination

Loads - Values

- Cases: 1to3 10to13

#	Case	Load type	List	Load values
1	1	self-weight	1to33 35to58 62to68 72to78	PZ Negative Factor=1.00
2	1	uniform load	54	PZ=-0.40(kN/m)
3	1	uniform load	66to68 76to78	PZ=-0.25(kN/m)
4	1	uniform load	59to61 69to71	PZ=-0.02(kN/m)
5	2	uniform load	66to68 76to78	PZ=-1.93(kN/m)
6	2	uniform load	54	PZ=-0.70(kN/m)
7	3	uniform load	55to57	PX=-0.01(kN/m)

Manual Combinations

- Cases: 10to13

Combinations	Name	Analysis type	Combination type	Case nature	Definition
10 (C)	SBC 1.2D+1.6L	Linear Combination	ULS	dead	1*1.20+(2+3)*1.60
11 (C)	SBC SLS D+L	Linear Combination	SLS	dead	(1+2+3)*1.00
12 (C)	SBC 1.4D	Linear Combination	ULS	dead	1*1.40
13 (C)	SBC 0.9D+1.6L (uplift)	Linear Combination	ULS	dead	1*0.90+3*1.60

Reactions ULS: global extremes

in the coordinate system: global - Cases: 1to3 10 12 13

	FX (kN)	FY (kN)	FZ (kN)	MX (kNm)	MY (kNm)	MZ (kNm)
MAX	0.46	0.40	4.33	0.00	0.00	0.00
Node	14	20	24	27	24	24
Case	10 (C)	10 (C)	10 (C)	10 (C)	2	10 (C)
MIN	-0.37	-0.58	-0.07	-0.00	-0.00	-0.00
Node	1	24	14	24	20	8
Case	10 (C)	10 (C)	3	10 (C)	10 (C)	10 (C)

Displacements SLS: global extremes

- Cases: 1to3 11

	UX (mm)	UY (mm)	UZ (mm)	RX (Rad)	RY (Rad)	RZ (Rad)
MAX	1.7	0.4	0.0	0.002	0.002	0.001
Node	56	49	19	38	24	32
Case	11 (C)	11 (C)	2	11 (C)	11 (C)	11 (C)
MIN	-1.0	-0.1	-2.8	-0.002	-0.005	-0.001
Node	26	56	35	32	55	38
Case	11 (C)	2	11 (C)	11 (C)	11 (C)	11 (C)

Member Forces ULS: envelope

- Cases: 1to3 10 12 13

Member	FX (kN)	FY (kN)	FZ (kN)	MX (kNm)	MY (kNm)	MZ (kNm)
1 / MAX	1.66	0.00	0.37	0.0	0.01	0.00
Node	1	1	1	1	2	2
Case	10 (C)	3	10 (C)	1	10 (C)	10 (C)
1 / MIN	0.07	-0.02	-0.01	-0.00	-0.00	-0.00
Node	1	1	1	1	2	2
Case	3	10 (C)	3	10 (C)	3	3
2 / MAX	1.75	0.00	-0.01	-0.00	0.00	0.00

Node	2	2	2	2	2	2
Case	10 (C)	13 (C)	3	3	3	13 (C)
2 / MIN	0.07	-0.00	-0.14	-0.00	-0.02	-0.00
Node	2	2	2	2	3	2
Case	3	10 (C)	10 (C)	10 (C)	10 (C)	10 (C)
3 / MAX	0.95	0.00	0.02	-0.00	0.02	0.00
Node	3	3	3	3	4	4
Case	10 (C)	13 (C)	10 (C)	3	10 (C)	10 (C)
3 / MIN	0.01	-0.00	-0.00	-0.00	-0.02	-0.00
Node	3	3	3	3	3	4
Case	3	10 (C)	3	10 (C)	10 (C)	13 (C)
4 / MAX	0.89	0.10	-0.00	0.00	0.02	0.04
Node	4	4	4	4	4	4
Case	10 (C)	10 (C)	3	12 (C)	10 (C)	10 (C)
4 / MIN	0.00	-0.00	-0.01	-0.00	-0.00	-0.09
Node	4	4	4	4	5	5
Case	3	3	10 (C)	3	3	10 (C)
5 / MAX	0.10	-0.00	0.86	0.01	0.00	0.00
Node	5	5	5	5	6	6
Case	10 (C)	3	10 (C)	10 (C)	3	10 (C)
5 / MIN	-0.00	-0.01	0.00	-0.00	-0.09	-0.00
Node	5	5	5	5	5	5
Case	3	10 (C)	3	3	10 (C)	3
6 / MAX	0.17	0.05	0.10	0.02	0.00	0.00
Node	6	6	6	6	7	7
Case	10 (C)	10 (C)	10 (C)	10 (C)	3	3
6 / MIN	0.00	-0.00	0.01	0.00	-0.03	-0.05
Node	6	6	6	6	6	7
Case	3	3	3	3	10 (C)	10 (C)
7 / MAX	1.46	0.00	0.10	0.00	0.00	0.00
Node	8	8	8	8	9	9
Case	10 (C)	2	10 (C)	3	10 (C)	13 (C)
7 / MIN	0.07	-0.00	-0.02	-0.00	-0.00	-0.00
Node	8	8	8	8	9	9
Case	3	13 (C)	13 (C)	10 (C)	13 (C)	2
8 / MAX	1.51	0.00	-0.01	0.00	0.00	0.00
Node	9	9	9	9	9	9
Case	10 (C)	10 (C)	3	10 (C)	3	10 (C)
8 / MIN	0.07	-0.00	-0.11	0.00	-0.02	-0.00
Node	9	9	9	9	10	9
Case	3	13 (C)	10 (C)	3	10 (C)	3
9 / MAX	0.85	0.00	0.02	0.00	0.02	0.00
Node	10	10	10	10	11	11
Case	10 (C)	2	10 (C)	10 (C)	10 (C)	12 (C)
9 / MIN	0.01	-0.00	-0.00	0.00	-0.01	-0.00
Node	10	10	10	10	10	11
Case	3	12 (C)	3	3	10 (C)	2
10 / MAX	0.80	0.00	-0.00	0.00	0.02	0.09
Node	11	11	11	11	11	12
Case	10 (C)	3	3	3	10 (C)	10 (C)
10 / MIN	0.00	-0.10	-0.01	-0.00	-0.00	-0.04
Node	11	11	11	11	12	11
Case	3	10 (C)	10 (C)	10 (C)	13 (C)	10 (C)
11 / MAX	0.61	-0.01	0.00	-0.00	-0.00	0.10
Node	12	12	12	12	13	13
Case	10 (C)	3	10 (C)	3	3	10 (C)
11 / MIN	0.00	-0.94	-0.00	-0.01	-0.02	-0.00
Node	12	12	12	12	12	12
Case	3	10 (C)	12 (C)	10 (C)	10 (C)	13 (C)
12 / MAX	0.46	0.03	-0.00	0.00	0.00	0.01
Node	14	14	14	14	14	15

Case	12 (C)	12 (C)	3	10 (C)	3	10 (C)
12 / MIN	-0.07	-0.19	-0.46	0.00	-0.01	-0.00
Node	14	14	14	14	15	15
Case	3	10 (C)	10 (C)	3	10 (C)	12 (C)
13 / MAX	0.40	0.08	0.05	0.02	0.00	0.02
Node	15	15	15	15	15	15
Case	12 (C)	10 (C)	10 (C)	10 (C)	3	10 (C)
13 / MIN	-0.07	-0.00	-0.00	-0.00	-0.03	0.00
Node	15	15	15	15	15	15
Case	3	3	3	3	10 (C)	3
14 / MAX	0.33	0.07	0.02	0.00	0.02	0.02
Node	16	16	16	16	17	16
Case	12 (C)	10 (C)	10 (C)	10 (C)	10 (C)	10 (C)
14 / MIN	-0.07	0.00	-0.00	-0.00	-0.01	-0.07
Node	16	16	16	16	16	17
Case	3	3	3	3	10 (C)	10 (C)
15 / MAX	-0.06	0.07	-0.00	-0.00	0.02	0.07
Node	17	17	17	17	17	17
Case	3	10 (C)	3	3	10 (C)	10 (C)
15 / MIN	-0.96	0.00	-0.03	-0.00	-0.01	-0.03
Node	18	17	17	17	18	18
Case	10 (C)	3	10 (C)	10 (C)	10 (C)	10 (C)
16 / MAX	-0.00	0.07	-0.01	-0.00	-0.00	-0.00
Node	18	18	18	18	18	18
Case	3	10 (C)	3	3	3	3
16 / MIN	-0.24	0.00	-0.19	-0.00	-0.02	-0.03
Node	19	18	18	18	19	19
Case	10 (C)	3	10 (C)	10 (C)	10 (C)	10 (C)
17 / MAX	4.20	0.40	0.12	0.00	0.12	0.00
Node	20	20	21	21	21	20
Case	10 (C)	10 (C)	10 (C)	3	10 (C)	10 (C)
17 / MIN	-0.00	0.00	-0.00	-0.01	-0.00	-0.01
Node	20	21	21	21	21	21
Case	3	3	3	10 (C)	3	10 (C)
18 / MAX	0.97	0.01	0.11	0.00	0.12	0.00
Node	21	21	21	21	22	21
Case	10 (C)	10 (C)	10 (C)	3	10 (C)	3
18 / MIN	-0.00	0.00	-0.00	-0.01	-0.00	-0.03
Node	21	21	21	21	22	22
Case	3	1	3	10 (C)	3	10 (C)
19 / MAX	0.92	0.03	0.15	0.00	0.20	-0.00
Node	22	22	22	22	23	22
Case	10 (C)	10 (C)	10 (C)	3	10 (C)	3
19 / MIN	-0.00	0.00	-0.00	-0.04	-0.00	-0.05
Node	22	22	22	22	23	23
Case	3	3	3	10 (C)	3	10 (C)
20 / MAX	4.33	0.00	0.15	0.01	0.14	0.02
Node	24	25	25	25	25	25
Case	10 (C)	12 (C)	10 (C)	10 (C)	10 (C)	10 (C)
20 / MIN	-0.00	-0.58	-0.00	-0.00	-0.00	-0.00
Node	24	24	25	25	25	25
Case	3	10 (C)	3	3	3	3
21 / MAX	0.86	0.00	0.14	0.01	0.15	0.02
Node	25	25	25	25	26	26
Case	10 (C)	12 (C)	10 (C)	10 (C)	10 (C)	10 (C)
21 / MIN	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00
Node	25	25	25	25	26	25
Case	3	2	3	3	3	3
22 / MAX	0.63	0.39	0.01	0.00	0.00	0.0
Node	27	27	27	27	28	27
Case	10 (C)	10 (C)	12 (C)	12 (C)	12 (C)	1

22 / MIN	-0.07	0.00	-0.20	0.0	-0.01	-0.01
Node	27	27	27	27	28	28
Case	3	3	10 (C)	2	10 (C)	10 (C)
23 / MAX	0.52	0.00	0.02	0.00	0.02	0.09
Node	28	28	28	28	29	29
Case	10 (C)	3	10 (C)	3	10 (C)	10 (C)
23 / MIN	-0.07	-0.09	-0.00	-0.00	-0.01	-0.04
Node	28	28	28	28	28	28
Case	3	10 (C)	3	10 (C)	10 (C)	10 (C)
24 / MAX	-0.04	-0.00	-0.00	0.00	0.02	0.04
Node	29	29	29	29	29	30
Case	1	3	3	10 (C)	10 (C)	10 (C)
24 / MIN	-0.76	-0.11	-0.02	0.00	-0.01	-0.10
Node	30	29	29	29	30	29
Case	10 (C)	10 (C)	10 (C)	3	10 (C)	10 (C)
25 / MAX	-0.00	-0.00	-0.01	0.00	-0.00	0.04
Node	30	30	30	30	30	31
Case	3	3	3	10 (C)	3	10 (C)
25 / MIN	-0.18	-0.10	-0.15	0.00	-0.01	0.00
Node	31	30	30	30	31	30
Case	10 (C)	10 (C)	10 (C)	3	10 (C)	3
26 / MAX	-0.00	-0.00	0.00	0.02	-0.00	0.01
Node	7	7	7	7	32	32
Case	1	3	3	10 (C)	3	10 (C)
26 / MIN	-0.13	-0.14	-0.06	0.00	-0.04	-0.05
Node	7	7	32	7	32	7
Case	10 (C)	10 (C)	10 (C)	3	10 (C)	10 (C)
27 / MAX	1.92	-0.00	0.27	0.04	0.12	0.04
Node	32	32	32	32	33	33
Case	10 (C)	3	10 (C)	10 (C)	10 (C)	10 (C)
27 / MIN	0.00	-0.08	0.00	0.00	-0.09	-0.02
Node	32	32	32	32	32	32
Case	3	10 (C)	3	3	10 (C)	10 (C)
28 / MAX	4.53	-0.01	0.50	0.06	0.13	0.07
Node	33	33	33	33	34	34
Case	10 (C)	3	10 (C)	10 (C)	10 (C)	10 (C)
28 / MIN	0.01	-0.12	0.00	0.00	0.00	0.00
Node	33	33	33	33	33	33
Case	3	10 (C)	3	3	3	3
29 / MAX	4.36	-0.00	0.08	-0.00	0.04	0.01
Node	34	34	34	34	35	35
Case	10 (C)	3	10 (C)	3	10 (C)	10 (C)
29 / MIN	0.00	-0.03	-0.00	-0.00	-0.01	-0.01
Node	34	34	34	34	34	34
Case	3	10 (C)	3	10 (C)	12 (C)	10 (C)
30 / MAX	4.48	0.01	-0.00	0.01	0.04	0.00
Node	35	35	35	35	35	35
Case	10 (C)	10 (C)	3	10 (C)	10 (C)	10 (C)
30 / MIN	0.00	0.00	-0.03	0.00	0.00	-0.01
Node	35	35	36	35	36	36
Case	3	3	10 (C)	3	3	10 (C)
31 / MAX	4.67	0.10	-0.00	0.00	0.09	0.06
Node	36	36	36	36	36	36
Case	10 (C)	10 (C)	3	3	10 (C)	10 (C)
31 / MIN	0.01	0.01	-0.48	-0.06	0.00	0.00
Node	36	36	37	36	37	37
Case	3	3	10 (C)	10 (C)	3	3
32 / MAX	4.06	0.09	-0.00	-0.00	0.13	0.05
Node	37	37	37	37	37	37
Case	10 (C)	10 (C)	3	3	10 (C)	10 (C)
32 / MIN	0.01	0.00	-0.26	-0.04	-0.07	-0.02

Node	37	37	38	37	38	38
Case	3	3	10 (C)	10 (C)	10 (C)	10 (C)
33 / MAX	0.77	0.07	-0.00	-0.00	0.10	0.00
Node	38	38	38	38	38	38
Case	10 (C)	10 (C)	3	3	10 (C)	12 (C)
33 / MIN	-0.00	0.00	-0.40	-0.04	-0.14	-0.04
Node	38	38	13	38	13	13
Case	3	3	10 (C)	10 (C)	10 (C)	10 (C)
35 / MAX	-0.01	-0.00	0.33	0.04	0.12	0.01
Node	6	6	6	6	39	39
Case	3	3	10 (C)	10 (C)	10 (C)	10 (C)
35 / MIN	-2.02	-0.02	0.00	0.00	-0.13	-0.01
Node	6	6	6	6	6	6
Case	10 (C)	10 (C)	3	3	10 (C)	10 (C)
36 / MAX	-0.02	0.02	0.04	0.01	0.04	0.02
Node	39	39	39	39	40	39
Case	3	10 (C)	10 (C)	10 (C)	10 (C)	10 (C)
36 / MIN	-4.63	0.00	0.00	-0.00	0.00	0.00
Node	39	39	39	39	39	40
Case	10 (C)	3	3	3	3	1
37 / MAX	-0.02	-0.00	0.07	0.00	0.06	0.01
Node	40	40	40	40	41	41
Case	3	3	10 (C)	3	10 (C)	10 (C)
37 / MIN	-4.75	-0.02	0.00	-0.00	0.00	-0.01
Node	40	40	40	40	40	40
Case	10 (C)	10 (C)	3	10 (C)	3	10 (C)
38 / MAX	-0.02	0.00	-0.00	-0.00	0.12	0.01
Node	41	41	41	41	41	42
Case	3	3	3	3	10 (C)	10 (C)
38 / MIN	-4.14	-0.01	-0.23	-0.02	-0.06	0.00
Node	41	41	42	41	42	42
Case	10 (C)	10 (C)	10 (C)	10 (C)	10 (C)	3
39 / MAX	-0.01	0.01	-0.00	-0.00	0.06	0.00
Node	42	42	42	42	42	42
Case	3	10 (C)	3	3	10 (C)	10 (C)
39 / MIN	-0.85	0.00	-0.15	-0.02	-0.08	-0.01
Node	42	42	12	42	12	12
Case	10 (C)	1	10 (C)	10 (C)	10 (C)	10 (C)
40 / MAX	2.07	0.06	-0.00	0.01	0.02	-0.00
Node	6	6	6	6	6	6
Case	10 (C)	10 (C)	3	10 (C)	10 (C)	3
40 / MIN	0.01	0.00	-0.17	-0.00	-0.05	-0.03
Node	6	6	32	6	32	32
Case	3	3	10 (C)	3	10 (C)	10 (C)
41 / MAX	2.59	-0.00	-0.00	0.02	0.10	0.00
Node	39	39	39	39	39	33
Case	10 (C)	3	3	10 (C)	10 (C)	10 (C)
41 / MIN	0.01	-0.04	-0.39	0.00	-0.06	-0.01
Node	39	39	33	39	33	39
Case	3	10 (C)	10 (C)	3	10 (C)	10 (C)
42 / MAX	0.11	0.04	-0.00	0.02	0.04	0.00
Node	40	40	40	40	40	40
Case	10 (C)	10 (C)	3	10 (C)	10 (C)	10 (C)
42 / MIN	-0.00	0.00	-0.10	0.00	-0.01	-0.01
Node	40	40	35	40	35	35
Case	3	3	10 (C)	3	12 (C)	10 (C)
43 / MAX	-0.00	-0.00	0.40	0.02	0.10	0.00
Node	41	41	41	41	37	37
Case	3	1	10 (C)	10 (C)	10 (C)	10 (C)
43 / MIN	-0.53	-0.01	0.00	0.00	-0.06	-0.00
Node	37	41	41	41	41	41

Case	10 (C)	10 (C)	3	3	10 (C)	10 (C)
44 / MAX	-0.01	-0.00	0.70	0.00	0.17	0.02
Node	42	42	42	42	38	38
Case	3	3	10 (C)	10 (C)	10 (C)	10 (C)
44 / MIN	-3.23	-0.02	0.00	0.00	-0.12	0.00
Node	38	42	42	42	42	42
Case	10 (C)	10 (C)	3	3	10 (C)	3
45 / MAX	-0.01	0.20	0.23	-0.00	0.06	0.06
Node	7	7	7	7	19	7
Case	3	10 (C)	10 (C)	3	10 (C)	10 (C)
45 / MIN	-0.10	0.01	0.00	-0.04	-0.07	-0.05
Node	7	7	7	7	7	19
Case	10 (C)	3	3	10 (C)	10 (C)	10 (C)
46 / MAX	-0.00	-0.01	0.21	0.04	0.05	0.05
Node	13	13	13	13	31	31
Case	3	3	10 (C)	10 (C)	10 (C)	10 (C)
46 / MIN	-0.07	-0.18	0.00	0.00	-0.06	-0.06
Node	13	13	13	13	13	13
Case	10 (C)	10 (C)	3	3	10 (C)	10 (C)
47 / MAX	0.82	-0.00	0.01	0.00	0.00	0.00
Node	3	3	3	3	18	18
Case	10 (C)	3	12 (C)	12 (C)	2	10 (C)
47 / MIN	0.06	-0.00	-0.01	-0.00	-0.00	-0.00
Node	3	3	18	3	3	3
Case	3	10 (C)	12 (C)	10 (C)	12 (C)	10 (C)
48 / MAX	0.67	0.00	0.01	0.00	0.00	0.00
Node	10	10	10	10	30	10
Case	10 (C)	10 (C)	12 (C)	10 (C)	2	10 (C)
48 / MIN	0.07	0.00	-0.01	-0.00	-0.00	-0.00
Node	10	10	30	10	10	30
Case	3	3	12 (C)	12 (C)	12 (C)	10 (C)
49 / MAX	0.00	0.00	0.01	0.03	0.00	0.06
Node	4	4	4	4	17	17
Case	3	3	10 (C)	10 (C)	10 (C)	10 (C)
49 / MIN	-0.03	-0.10	-0.01	0.00	-0.00	-0.00
Node	4	4	17	4	4	4
Case	10 (C)	10 (C)	12 (C)	3	13 (C)	10 (C)
50 / MAX	0.00	0.10	0.01	0.00	0.00	0.00
Node	11	11	11	11	29	11
Case	3	10 (C)	10 (C)	3	10 (C)	10 (C)
50 / MIN	-0.03	-0.00	-0.01	-0.04	-0.00	-0.05
Node	11	11	29	11	29	29
Case	10 (C)	3	12 (C)	10 (C)	12 (C)	10 (C)
51 / MAX	-0.00	0.00	0.00	0.00	0.02	0.01
Node	2	2	2	2	2	15
Case	3	3	3	13 (C)	10 (C)	10 (C)
51 / MIN	-0.50	-0.02	-0.11	-0.00	-0.04	-0.00
Node	2	2	15	2	15	15
Case	10 (C)	10 (C)	10 (C)	10 (C)	10 (C)	3
52 / MAX	-0.00	-0.00	0.00	0.00	0.02	0.00
Node	9	9	9	9	9	28
Case	1	3	3	10 (C)	10 (C)	3
52 / MIN	-0.21	-0.00	-0.07	-0.00	-0.02	-0.00
Node	9	9	28	9	28	9
Case	10 (C)	10 (C)	10 (C)	3	10 (C)	10 (C)
53 / MAX	-0.00	0.00	0.03	0.02	-0.00	0.00
Node	28	28	15	28	28	15
Case	3	10 (C)	10 (C)	10 (C)	3	3
53 / MIN	-0.48	-0.01	-0.04	-0.02	-0.02	-0.01
Node	28	15	28	15	28	15
Case	10 (C)	10 (C)	10 (C)	10 (C)	10 (C)	10 (C)

54 / MAX	0.00	0.07	1.04	0.00	-0.00	0.06
Node	29	17	17	29	29	17
Case	3	10 (C)	10 (C)	12 (C)	3	10 (C)
54 / MIN	-0.12	-0.07	-1.19	-0.00	-0.24	-0.00
Node	29	29	29	29	29	29
Case	10 (C)	10 (C)	10 (C)	3	10 (C)	3
55 / MAX	0.27	0.00	0.01	0.04	0.01	0.06
Node	19	23	19	19	19	23
Case	10 (C)	3	12 (C)	10 (C)	10 (C)	10 (C)
55 / MIN	0.01	-0.09	-0.06	0.00	-0.05	-0.06
Node	19	19	23	19	23	19
Case	3	10 (C)	10 (C)	3	10 (C)	10 (C)
56 / MAX	0.47	0.01	0.02	0.01	0.03	-0.00
Node	23	26	23	23	23	26
Case	10 (C)	13 (C)	12 (C)	10 (C)	10 (C)	1
56 / MIN	0.02	-0.01	-0.04	0.00	-0.00	-0.00
Node	23	23	26	23	26	23
Case	3	10 (C)	10 (C)	3	12 (C)	10 (C)
57 / MAX	0.28	0.08	0.04	-0.00	0.00	0.05
Node	26	31	26	26	31	26
Case	10 (C)	10 (C)	10 (C)	3	2	10 (C)
57 / MIN	0.01	-0.00	-0.02	-0.04	-0.03	-0.05
Node	26	26	31	26	26	31
Case	3	3	12 (C)	10 (C)	10 (C)	10 (C)
58 / MAX	0.07	0.04	0.02	0.01	0.00	0.03
Node	16	16	16	16	16	16
Case	10 (C)	10 (C)	12 (C)	10 (C)	2	10 (C)
58 / MIN	-0.04	-0.00	-0.03	0.00	-0.03	-0.08
Node	22	16	22	16	22	22
Case	12 (C)	3	10 (C)	3	10 (C)	10 (C)
59 / MAX	0.09	0.17	0.41	-0.00	0.09	0.08
Node	34	34	34	34	43	34
Case	10 (C)	10 (C)	10 (C)	3	10 (C)	10 (C)
59 / MIN	0.00	0.01	0.00	-0.14	0.00	0.00
Node	34	34	34	34	34	43
Case	3	3	3	10 (C)	3	3
60 / MAX	0.09	0.17	0.00	-0.00	0.09	0.07
Node	43	43	43	43	43	43
Case	10 (C)	10 (C)	3	3	10 (C)	10 (C)
60 / MIN	0.00	0.01	-0.35	-0.14	-0.01	0.00
Node	43	43	44	43	44	44
Case	3	3	10 (C)	10 (C)	12 (C)	3
61 / MAX	0.08	0.17	0.00	-0.00	0.02	0.03
Node	44	44	44	44	44	44
Case	10 (C)	10 (C)	3	3	10 (C)	10 (C)
61 / MIN	0.00	0.01	-0.85	-0.13	-0.24	-0.02
Node	44	44	23	44	23	23
Case	3	3	10 (C)	10 (C)	10 (C)	10 (C)
62 / MAX	0.00	0.00	0.00	0.00	0.00	0.00
Node	43	43	43	43	43	43
Case	3	3	10 (C)	3	3	3
62 / MIN	-0.75	-0.00	-0.00	-0.00	-0.00	-0.00
Node	43	43	43	43	43	43
Case	10 (C)	10 (C)	3	10 (C)	10 (C)	10 (C)
63 / MAX	0.00	0.00	0.00	0.00	0.00	0.00
Node	45	45	45	45	46	46
Case	3	3	10 (C)	3	10 (C)	10 (C)
63 / MIN	-0.75	-0.00	-0.00	-0.00	-0.00	-0.00
Node	45	45	45	45	45	45
Case	10 (C)	10 (C)	3	10 (C)	10 (C)	10 (C)
64 / MAX	-0.00	0.00	0.00	0.00	0.00	-0.00

Node	44	44	44	44	44	44
Case	3	10 (C)	10 (C)	3	3	3
64 / MIN	-0.50	0.00	-0.00	-0.00	-0.00	-0.00
Node	44	44	44	44	44	47
Case	10 (C)	3	3	10 (C)	10 (C)	10 (C)
65 / MAX	-0.00	0.00	0.00	0.00	0.00	-0.00
Node	47	47	47	47	48	47
Case	3	10 (C)	10 (C)	3	10 (C)	3
65 / MIN	-0.50	0.00	-0.00	-0.00	-0.00	-0.00
Node	47	47	47	47	47	48
Case	10 (C)	3	3	10 (C)	10 (C)	10 (C)
66 / MAX	0.01	0.00	0.92	-0.00	0.00	0.00
Node	21	21	21	21	48	21
Case	10 (C)	3	10 (C)	3	10 (C)	3
66 / MIN	-0.00	-0.00	-0.13	-0.00	-0.11	-0.00
Node	21	21	48	21	21	21
Case	3	10 (C)	10 (C)	10 (C)	10 (C)	10 (C)
67 / MAX	0.00	0.00	0.35	0.00	0.00	0.00
Node	48	48	48	48	48	48
Case	10 (C)	3	10 (C)	10 (C)	12 (C)	3
67 / MIN	-0.00	-0.00	-0.45	-0.00	-0.01	-0.00
Node	48	48	46	48	46	48
Case	3	10 (C)	10 (C)	3	10 (C)	10 (C)
68 / MAX	0.0	0.00	0.28	0.00	-0.00	0.00
Node	46	46	46	46	49	46
Case	2	3	10 (C)	3	3	3
68 / MIN	-0.00	-0.00	0.00	-0.00	-0.01	-0.00
Node	46	46	46	46	46	46
Case	10 (C)	10 (C)	3	10 (C)	10 (C)	10 (C)
69 / MAX	0.09	-0.01	0.45	0.06	0.10	-0.00
Node	36	36	36	36	50	50
Case	10 (C)	3	10 (C)	10 (C)	10 (C)	3
69 / MIN	0.01	-0.19	0.00	0.00	0.00	-0.06
Node	36	36	36	36	36	36
Case	3	10 (C)	3	3	3	10 (C)
70 / MAX	0.09	-0.01	0.00	0.06	0.10	-0.00
Node	50	50	50	50	50	51
Case	10 (C)	3	3	10 (C)	10 (C)	3
70 / MIN	0.01	-0.19	-0.33	0.00	-0.01	-0.05
Node	50	50	51	50	51	50
Case	3	10 (C)	10 (C)	3	12 (C)	10 (C)
71 / MAX	0.09	-0.01	0.00	0.06	0.03	0.05
Node	51	51	51	51	51	26
Case	10 (C)	3	3	10 (C)	10 (C)	10 (C)
71 / MIN	0.01	-0.18	-0.76	0.00	-0.20	-0.01
Node	51	51	26	51	26	51
Case	3	10 (C)	10 (C)	3	10 (C)	10 (C)
72 / MAX	0.00	0.00	0.00	0.00	0.00	0.00
Node	50	50	50	50	50	50
Case	3	10 (C)	10 (C)	10 (C)	3	10 (C)
72 / MIN	-0.77	-0.00	-0.00	-0.00	-0.00	-0.00
Node	50	50	50	50	50	50
Case	10 (C)	3	3	3	10 (C)	3
73 / MAX	0.00	0.00	0.00	0.00	0.00	0.00
Node	52	52	52	52	53	52
Case	3	10 (C)	10 (C)	10 (C)	10 (C)	10 (C)
73 / MIN	-0.77	-0.00	-0.00	-0.00	-0.00	-0.00
Node	52	52	52	52	52	53
Case	10 (C)	3	3	3	10 (C)	10 (C)
74 / MAX	-0.00	-0.00	0.00	0.00	0.00	0.00
Node	51	51	51	51	51	54

Case	3	3	10 (C)	10 (C)	3	10 (C)
74 / MIN	-0.42	-0.00	-0.00	-0.00	-0.00	-0.00
Node	51	51	51	51	51	51
Case	10 (C)	10 (C)	3	3	10 (C)	3
75 / MAX	0.02	-0.00	0.00	0.00	0.00	0.00
Node	55	54	54	54	55	55
Case	12 (C)	3	10 (C)	10 (C)	10 (C)	10 (C)
75 / MIN	-0.42	-0.00	-0.00	-0.00	-0.00	-0.00
Node	54	54	54	54	54	54
Case	10 (C)	10 (C)	3	3	10 (C)	3
76 / MAX	0.01	-0.00	0.98	0.00	0.01	0.00
Node	25	25	25	25	55	55
Case	10 (C)	3	10 (C)	10 (C)	10 (C)	10 (C)
76 / MIN	-0.00	-0.00	-0.07	0.00	-0.13	-0.00
Node	25	25	55	25	25	25
Case	3	10 (C)	2	3	10 (C)	13 (C)
77 / MAX	0.00	0.00	0.34	0.00	0.01	0.00
Node	55	55	55	55	55	55
Case	10 (C)	10 (C)	10 (C)	3	10 (C)	10 (C)
77 / MIN	-0.00	-0.00	-0.47	-0.00	-0.01	-0.00
Node	55	55	53	55	53	55
Case	3	3	10 (C)	10 (C)	10 (C)	3
78 / MAX	0.00	0.00	0.28	0.00	0.00	0.00
Node	53	53	53	53	56	53
Case	10 (C)	3	10 (C)	10 (C)	12 (C)	3
78 / MIN	-0.00	-0.00	0.00	-0.00	-0.01	-0.00
Node	53	53	53	53	53	53
Case	3	10 (C)	3	3	10 (C)	10 (C)

Stories - Displacements

Case/Story	Displacement and drift for all nodes at story level						Extreme story displacement	
	UX (mm)	UY (mm)	dr UX (mm)	dr UY (mm)	d UX	d UY	Max UX (mm)	Max UY (mm)

Stories - Reduced forces

Case/Story	G (x,y,z) (m)	FX (kN)	FY (kN)	MZ (kNm)	FX to columns (kN)	FX to walls (kN)	FY to columns (kN)	FY to wal (kN)